SDC HACKATHON 2026

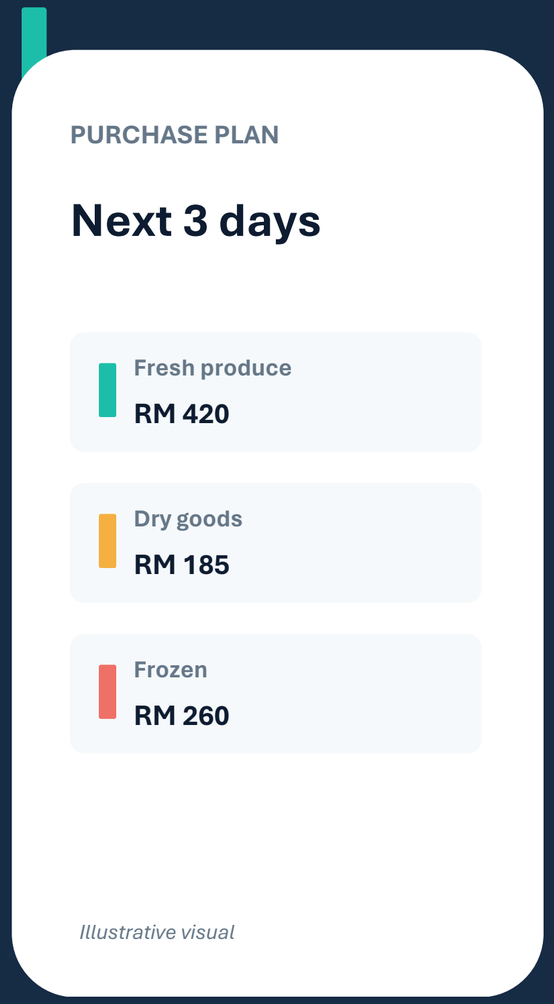
RestockAI

Smarter purchasing.
Less waste. More control.

AI-assisted inventory and purchase planning for restaurants & cafés

DOMAIN 2 · ORBYTE SMART SME CHALLENGE

FINAL PROJECT PRESENTATION



Restaurant purchasing is a daily balancing act

The cost of getting the order wrong shows up in time, waste and service.

Too much manual work

Staff combine sales history, recipes, stock counts and supplier prices by hand. Planning can take close to two days.

Stockouts hurt service

Buying too little can mean missing menu items, lost sales and unhappy customers.

Overbuying creates waste

Perishables may expire before they are used, while pack sizes and minimum orders add complexity.

THE OPPORTUNITY

Turn scattered operational data into a clear, explainable purchase plan.

SAY Lead with the operational problem, not the technology.

From sales history to a supplier-ready plan

RestockIQ connects the data restaurants already have, then keeps people in control.

01

Import sales

CSV · Excel · PDF

→

02

Forecast demand

Calendar + history

→

03

Check stock

Expiry + shelf life

→

04

Plan purchases

Packs + cost

→

05

Review & approve

Human confirmation

DESIGN PRINCIPLE

Use AI for messy input and helpful explanations — keep core arithmetic deterministic and auditable.

SAY Explain that AI supports the workflow; it does not make the purchasing decision alone.

Purchase Plan: the decision screen



DISH	TODAY 21 Oct	THU 22 Oct	TOTAL
Sambal Chicken Rice 香辣鸡丁饭	15	14	29
Nasi Goreng Tomyam 东炎炒饭	11	10	21
Waffle	10	10	20
Green Tea 绿茶 (Cold)	10	10	20
Nasi Goreng Kampung 马来家乡炒饭	8	7	15
Ayam Gepuk	7	8	15
Fried Chicken Chop with Rice 炸鸡扒饭	7	7	14
Nasi Goreng Pattaya 坐包饭	7	7	13
Tomyam Ayam Soup	6	6	12
Milo 美禄 (Cold)	5	5	10
Orange Juice 橙汁	5	5	10

What to point out

Demand forecast and its drivers

Why the quantity?

Need + buffer − usable stock

Can staff trust it?

Each line has a “Why?” explanation

SAY Walk through one ingredient line from forecast need to suggested packs.

RestockAI

First-time setup · Ah Hock

Let's set up Kopitiam Demo (AIMST)

1 Country 2 Sales history 3 Your stock 4 Recipes 5 Calendar (optional)

Where is your business?

Sets prices to MYR and adds Malaysian public holidays to your calendar.

- (•) **Malaysia**
  Prices in MYR (RM). Malaysian public holidays are added to your calendar.
- ( ) Other countries coming soon

Continue

Help

A simple architecture built for a practical SME workflow

A web app, a Python decision engine and a database — with AI used only where it helps.

FRONT END · React 18 + TypeScript + Vite + Tailwind + Recharts

Dashboards · Inventory · Sales data · Plan purchases · Orders · Help chat

↓

BACK END · Python + FastAPI

Authenticated API · validation · role permissions · activity log

↓

DECISION ENGINE · pandas + NumPy

Demand forecast · recipe requirements · backtested buffers · shelf-life simulation

↓

DATA + AI SERVICES

SQLAlchemy → SQLite locally / PostgreSQL in deployment · LLM for PDF extraction and help

SAY The forecast and purchasing maths run in Python, not inside the chatbot.

Forecast demand using context, not a flat average

Why this design: restaurant demand changes by weekday, season and local events.

Learn normal patterns

Use daily sales history to learn weekday effects and each menu item’s share of demand.

Adjust for calendar context

Public holidays, local events and business periods influence demand; effects are softened when examples are scarce.

React to recent changes

Recent trading strength helps capture a busy or quiet spell without hard-coding a growth assumption.

FORECAST IDEA

Expected dishes = baseline level × weekday pattern × calendar effect × recent strength

Cold start: with less than 8 weeks of history, use a simpler fallback and flag it.

SAY Mention that the model exposes its drivers so a person can understand why demand moved.

Buy enough — without ignoring expiry

The purchasing engine simulates stock day by day instead of treating all inventory as equally usable.

TODAY

Remove expired stock

EACH DAY

Consume oldest usable stock

WHEN SHORT

Calculate need + safety buffer

SUPPLIER PACKS

Round up to whole packs

Perishables may need multiple trips

A fresh batch is bought only for the window it can cover before expiry. Leftovers carry forward only while still usable.

Cost and supplier constraints stay visible

Plans are grouped by supplier, subtotalled, and flagged when below a supplier’s minimum order.

WHY IT MATTERS

Less avoidable waste, fewer stockouts, and a purchase list staff can explain.

SAY Use an example like eggs or milk: expiry changes what counts as usable stock.

AI handles unstructured input — people confirm the result

The app deliberately keeps AI away from the core demand and purchasing arithmetic.

1 · PDF sales extraction

A user uploads a sales PDF. The AI extracts rows into structured fields and notes uncertainty. The result stays a draft until a person checks and confirms the preview.

2 · Help chatbot

Users ask how the app works. The assistant draws on curated help content and plain-language guidance; it does not read live stock/sales or change records.

GUARDRAIL

AI output is not silently saved. Review first, then confirm.

CSV and Excel files use structured parsing; PDF reading requires the configured AI service.

SAY Call it AI-powered PDF extraction; avoid claiming a separate OCR pipeline unless implemented.

Chosen for speed, clarity and maintainability

A lightweight stack keeps the prototype understandable and deployable.

FRONT END

React 18

TypeScript · Vite · Tailwind CSS · Recharts

BACK END

FastAPI / Python

REST API · role checks · request validation

DECISION ENGINE

pandas + NumPy

Forecasting · backtesting · stock simulation

DATA

SQLAlchemy

SQLite locally · PostgreSQL in deployment

AI SERVICES

LLM integration

PDF sales extraction · help chatbot

DELIVERY

Web application

FastAPI serves API and built front end

SAY The LLM is an assistive service; core calculations are deterministic Python functions.

Recommendations are not orders until a person approves

A clear status flow reduces accidental actions and keeps responsibility with the team.

1
Generate
Manager creates a plan

→

2
Edit & submit
Review quantities and notes

→

3
Approve / reject
Owner makes the decision

→

4
Contact supplier
Use WhatsApp or phone

→

5
Mark received
Stock is updated on receipt

WHY THIS CHOICE?

Opening WhatsApp does not prove an order was sent. Approval and receipt require explicit human action.

SAY Emphasize the system supports decisions; it does not pretend to send or confirm orders.

10 · LIVE DEMO

Show the decision.
Explain the why.

THANK YOU

A simple demo path for the panel

01	**Open Purchase Plan**	Set a 3-day horizon and generate.
02	**Explain one item**	Point to forecast need, safety buffer, usable stock and pack rounding.
03	**Show expiry awareness**	Explain why expired stock is excluded and why perishables may be split into trips.
04	**Show the human check**	Edit or review the plan, then explain submit → approve → contact supplier → receive.
05	**Show AI in context**	Upload a sample PDF to preview extracted rows, or ask the help chatbot a how-to question.